

# ISHAN AVASTHI

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## EDUCATION

**Birla Institute of Technology And Science, Pilani (BITS Pilani)**

B.Sc. (Hons.) in Computer Science

*July 2023 – Present*

*CGPA - 8.6*

**Scaler School of Technology, Bangalore**

Undergraduate Program in Computer Science

*July 2023 – Present*

*CGR - 7.15*

## PROJECTS

**KV-Cache: High-Performance In-Memory Cache Server** | [View Project](#)

- **Concurrent Server Design:** Built a key-value cache server from scratch on raw TCP with Python **asyncio** non-blocking I/O, splitting network, protocol, and storage into separate layers with a custom wire-protocol parser, **O(1)** operations, **LRU eviction**, and TTL expiry.
- **Performance Benchmarking:** Sustained **13,348 requests/second** across **100 concurrent connections** over **100,000 requests**, at **p95 latency 8.1 ms**, **p99 9.6 ms**, and a **0% error rate**, measured on **AWS EC2**.
- **Testing & Deployment:** Covered store, eviction, TTL, protocol, and end-to-end server paths with a **pytest** suite, and shipped a **Docker** image with scripted EC2 provisioning, deployment, and load-test runs.

**AlphaDesk: Full-Stack Equity Research Platform** | [View Project](#) | [Live Demo](#)

Web application that automates Indian equity research end to end, from market scanning to a risk-checked report a user approves before it is saved.

- **Full-Stack Architecture:** Built a **Next.js** frontend against a **FastAPI REST** backend, streaming live progress to the browser over **Server-Sent Events**, persisting shareable analyses, and tracing every run with **LangSmith**.
- **API Integration & Auth:** Integrated an external market-data service over **OAuth 2.0 with PKCE** for live NSE prices, and built a document retrieval layer over filing PDFs using **ChromaDB** and local ONNX embeddings.
- **Backend Services:** Engineered a **5-stage LangGraph pipeline** (Scanner, Research, Analyst, RiskManager, Execution) sharing one **Pydantic** schema, with conditional routing and checkpointing so a failed run resumes instead of restarting.

**EventHive: Event Discovery & Ticket Booking Platform** | [View Project](#)

Built with one teammate: a platform where hosts publish events and attendees browse and book tickets, with a shared REST backend serving a mobile client.

- **REST API & Data Modelling:** Built an **Express.js** API on **MongoDB Atlas** with **Mongoose** schemas for users, events, and bookings, using document references and population to resolve host and attendee data in a single query.
- **Authentication & Validation:** Implemented **JWT** middleware guarding private routes, **bcrypt** password hashing, and **Google OAuth** sign-in, plus **express-validator** schema checks and **Helmet** and CORS hardening on every request.
- **Booking Flow & Client:** Modelled ticket inventory, pricing, and booking-status transitions with a payment-order integration, consumed by a **React Native (Expo)** client using **Context API** auth and a shared API service layer; backend deployed on **Render**.

**ContextLens: Chrome MV3 Extension & Postgres-Backed API** | [View Project](#)

**ClinicFlow: Real-Time AI Voice Receptionist** | [View Project](#) | [Demo Video](#)

## ACHIEVEMENTS

- Cleared the online coding round and got selected for the Amazon ML Summer School. [View Certificate](#)

## TECHNICAL SKILLS

- **Frontend:** JavaScript, TypeScript, React, Next.js, React Native, HTML, CSS, Tailwind CSS, Responsive UI
- **Backend & APIs:** Node.js, Express.js, REST API Design, FastAPI, Python, Java, JWT & OAuth 2.0, Server-Sent Events
- **Databases:** MongoDB (Mongoose, Atlas), SQL (PostgreSQL, SQLite), Vector Databases (ChromaDB)
- **Cloud & Tooling:** Git, GitHub Actions, Docker, AWS EC2, Vercel, Render, CI/CD, Pytest, Playwright, Load Testing
- **AI Engineering:** LLMs, RAG, Agentic Systems, LangGraph, MCP, Tool Calling, Structured Outputs
- **Relevant Coursework:** Data Structures & Algorithms, OOP, DBMS, OS, AI, Machine Learning